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Computerized cell tracking: Current methods, tools and challenges

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ABSTRACT

In developmental biology, knowledge of cell structure and their (morpho) dynamic behavior, leads to a comprehensive understanding of their conducts and the mechanisms in which they participate. This knowledge is a decisive factor in biological research and also in all drug development steps, medicinal or preventive therapies. Experimental cell analysis is hard, expensive, and time-consuming. To overcome these difficulties, in recent years, several computational object tracking methods, software system and packages have been developed in cell sciences that bring together different disciplines and branches of technologies.

Object tracking is the process of locating and monitoring specific object and its behavior in sequential images. In this paper, a comprehensive review on object tracking stages and computational methods that are utilized in terms of cell tracking has been organized. Besides, the available software packages and toolkits, challenges, and their solution in time lapse microscopy images in this scope were reviewed. The aim of describing computational cell tracking methods and tools is that biologist and cell scientists might take advantage of these computational techniques to find another method to gain complementary information for their question of interest.

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1. Introduction

The cell is the smallest body structural unit of all living organisms. In developmental biology realization of the growth, alignment, and partition of single cells in all morphogenesis of organs is one of the big challenges (Mankowski et al., 2014; Maška et al., 2014; Majumdar et al., 2019). Understanding the underlying mechanisms of cellular behavior is a decisive factor for medicinal or preventive therapies. For example, many diseases are due to an abnormality in the cells (e.g. cell mobility, cell differentiation, cell deviation, cell proliferation, etc.) (Sacan et al., 2008a). In recent years, various studies have been conducted on the cell functional, structural behavior, and complex biological processes such as cell

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migration (Tosi and Campbell, 2019), cell cycle (Spoerri et al., 2017), cell differentiation (Rapoport et al., 2011), components and location (Mankowski et al., 2014), self-assembly (Dou et al., 2019), gene regularization (Lopes et al., 2019), signal transduction (Kusumi et al., 2014), and active trafficking (Joensuu et al., 2017).

Cell events, such as membrane dynamics (Hatakeyama et al., 2017), cell death (Jiang and Poon, 2019), occlusion, development, rapid/suddenly motions, dynamics of cell motion (e.g. velocity, directionality), the morphological changes, unstable cell–cell linkages, decampment conduct, and cell behaviors, including the quantification of migration (Majumdar et al., 2019), mitosis (Moura and Conde, 2019) , and also the rebuilding of cell lineages (mother–daughter relations) (Shrier et al., 2004), need to be accurately captured.

Since these events are hard, time consuming and require expertise, pose many questions and problems, therefore they are suited to computational object tacking approaches (Li et al., 2008a). In addition, there are some challenges in manual analyzing of cell images, including (i) the large size of a single cell image sequence, (ii) diagnosing of a disease depends on long time, laborious work, the level of experience of the human operator and (iii) is almost non-reproducible.

Although, many efforts have been applied to improve tracking, but the small size of cells and their organelles raise numerous complicated problems that are appropriate to computational object tacking approaches (Nketia et al., 2017).

An object tracking study is the tracking or finding of an object of interest in images by using of live cell fluorescence microscopy images and computational algorithms (Dupont et al., 2013; Meijering et al., 2009a). A computational object tracking is applied to the sequential images to track cells events over time and extract signals from each object. As a result, using computational object tracking can notably improve the reproducibility, efficiency, and accuracy of the studies on cells. Moreover, due to the ability of these methods to visualize molecular activities in real time, they are a key to disease pattern detection and can be helpful in decision making for example research in pathology for diagnosis of a large number of diseases including most cancers.

Object tracking as a subfield of computer vision, has been engaged broadly for tracking of cells and subcellular particles in sequential images (Hatakeyama et al., 2017; Meijering et al., 2009b). Based on the literature, there are three stages for object tracking (Samdurkar et al., 2017a; Parekh et al., 2014; Mean SLDUA, 0000), including: object detection, object classification and object tracking. (Fig. 1)

Computational object tracking approaches are simple, cost and time effective, and do not require sophisticated instruments. This capacity is important for several fields such as genomics and proteomics, stem cell researches, drug discovery, and tissue engineering.

In this review, we provide a comprehensive overview of the computational object tracking, their existing methods, available tools, and different studies conducted in the field of object tracking in experimental biological imaging. There is complete focus on analysis and tracking of objects in time-lapse microscopy images. In this review, all available DB and literature searched to cover all the related necessary investigation for a fairly long period of time about three decades ago.

In the following sections, first, an overview of cell tracking stages, methods, and their advantages and disadvantages is described. Next, the computational tools in two available points intracellular and extracellular are elucidated. Some related works in terms of cell tracking approaches are also illustrated. Finally, a discussion of the challenges ahead and future researches is presented.

2. Cell tracking methods:

Computational object tracking is the process of locating a moving object in sequential images.

In cell approaches, computational object tracking can be applied for a various purposes. Table 1 summaries various application of object tracking in live cell.

Cell tracking methods can be classified into three branches containing tracking by detection, tracking by model evaluation, and tracking by filtering. This classification is applicable for both two-dimensional (2D) and three-dimensional (3D) data (Masuzzo et al., 2016).

2.1. Tracking by detection (detection based approach)

In these approaches, objects are first detected from the background based on intensity, texture, or gradient features (Al-Kofahi et al., 2006). Then, objects are matched between consecutive frames, by optimizing certain probabilistic objective functions. The object detection stage is achieved by a process of image segmentation. Segmentation assigns different labels to every pixel in each image, in order that pixels with the same label have particular properties (Masuzzo et al., 2016; Boukari and Makrogiannis, 2018a).

In recent years, several methods have been developed for image segmentation. Most of these methods use only 2D images. However, some of them are extended to 3D, and are furthermore intricate (Masuzzo et al., 2016). The main advantages of detection-based tracking approaches are: working on lower imaging frequencies, advanced data association techniques for tracking, and separation the segmentation and tracking tasks. Therefore, various segmentation methods can be applied with a single tracking algorithm (Boukari and Makrogiannis, 2018b). However, in some problems, tracking can be problematic (e.g. mitosis tracking) in due to the inability to discriminate between multiple touching cells, and consequently segmentation errors rate increase with increasing cell density.

Ong et al. (1996) conducted a review of image segmentation and categorized it into four approaches, including Thresholding, region growing, edge detection, and pattern matching. In the following paragraphs, we highlighted the most common segmentation methods.

Thresholding: One of the most predominant techniques for image segmentation is Thresholding. The Thresholding creates a binary image as an output which one condition will display the foreground objects, and complementary condition will match to the background (Sezgin and Sankur, 2004). Therefore, Thresholding is efficient and applicable to images with a high-contrast uniform background in which objects are homogeneous (Luo et al., 1998). However, there are several factors that intricate the Thresholding, such as impermanent and mutually noise, circumfused brightness, the gathering gray levels inside the object and its background, insufficient contrast, and the size of object disproportionate with the scene. Eventually, the shortage of object measures to assess the performance of different algorithms of Thresholding, and the difficulty of extended evaluation in a task directed environment, have been other restrictions (Sezgin and Sankur, 2004; Kong et al., 2015a). In practice, just Thresholding generates insufficient segmentation outcomes. Commonly, the most of cell segmentation techniques use Thresholding as a basic level in pipeline (Meijering, 2012).

Edge detection: This technique is based on the assumption that one edge in an image is belonged to an extreme change in image intensity (Gonzalez and Wintz, 1977). Edge detection often supplies effective outcomes for images with “good boundaries”

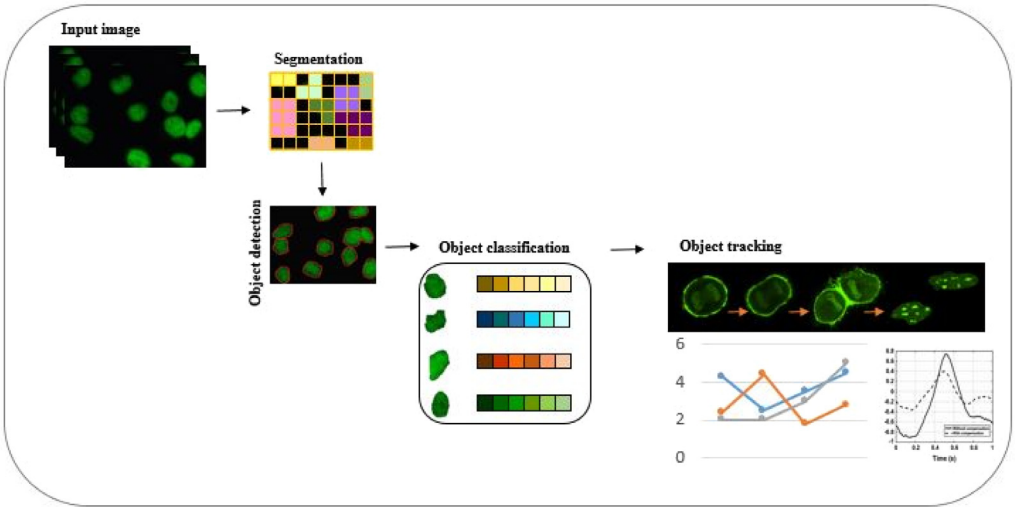


Fig. 1. The computational object tracking process: **(I) object detection:** objects are recognized in sequential images and their pixels are clustered. This step starts by segmenting the object in all frames. Some object detection techniques are frame differing, optical flow, background subtraction, segmentation, and point detectors. **(II) Object classification:** objects are classified according to their shape, motion, color and behavior. Shape-based, motion-based, color-based, and behavior-based classification are some object classification techniques. **(III) Object tracking:** a trajectory of an object is generated by locating its position in every frame of the sequential images in each moment.

Table 1
Applications of the object tracking to investigate live cells in terms of structure and culture.

Application	Description	Reference
Intracellular dynamic tracking	Tracking the intracellular organs and vesicles to understand cellular dynamics.	Kalaidzidis (2007b) and Moogk et al. (2007)
Particle tracking (PT)	Tracking the motion or behavior of particle/particles in an environment.	Rust et al. (2011), Celler et al. (2013), Feng et al. (2011), van der Schaar et al. (2008), Cognet et al. (2014), Bacher et al. (2004) and Gardini et al. (2015)
Molecule tracking (MT)	The class of non-invasive techniques for direct observation of the molecules space as a function of time.	Joensuu et al. (2016), MacGillavry and Blanpied (2013), Kusumi et al. (2005), Appelhans and Busch (2017), Bosch et al. (2014), Mehta et al. (2016), Ma et al. (2017), Kwakowsky et al. (2013) and Ram et al. (2012)
Protein tracking (PT)	Understanding the place and dynamics of protein pairs interacting or proteins position.	Courty and Dahan (2013) and Saurabh et al. (2017)
Cell tracking	Spatial studies of living cells motility that can include various concepts.	Kwak et al. (2010), Maglica et al. (2015), Piltti et al. (2017), Gómez-Villafuertes et al. (2017), Falconnet et al. (2011), Gahlmann and Moerner (2014), Hoornweg et al. (2016), Wen et al. (2017) and Youssef et al. (2011)
Tracking normal and malignant cell cycle	Tracking the behavior of cells and finding cells that are defective in their life-cycle; most used to find cancer cells.	Pineda et al. (2016) and Zerjatke et al. (2017)
Tracking of cell death	Studying tumor growth and treatment response of malignancies.	Skommer et al. (2010) and de et al. (2005)
Spot tracking	Cover a wide range of studies at the MRI, DNA level and translation dynamic.	Gardini et al. (2015), Asaithamby and Chen (2011), Rino et al. (2016), Li et al. (2016), Georgescu et al. (2015) and Thompson et al. (2010)
Cell behavior tracking	Includes tracking of cellular behaviors that cause morphogenesis, such as division, migration, or death.	Puliafito et al. (2012), Mukewar and Wang (2004), Huh (2013) and Bise et al. (2011)

meaning that intensity of changes is severe through the edge and homogeneous throughout the edge. Although, compared to Thresholding methods, techniques of edge-detection are stronger but they miscarry on low level of contrast images (Masuzzo et al., 2016). Edge detection in 2D is able to enlarge to 3D surface detection. In the time-lapse fluorescence microscopy images, the light diffracting neighboring fluorescent objects are able to cause blurriness at the edge of the object (Luo et al., 1998). One of the most restrictions of both edge-detection and Thresholding techniques is that, they are unable to recognize two objects with the same intensity and boundary (e.g. when cells are next to each other or overlap). This restriction is overcome by watershed technique, which the images are considered as like a topographic relief that the pixel gray levels are explicated as their altitude. Methods based on the watershed segmentation have higher performance at differentiating clumped objects, but their utility decreased into such low contrast conditions (specifically when local invisibility of cell membrane occurs) (Masuzzo et al., 2016). Since, this strategy can cause over-segmentation, as a result, specific pre- and post-processing methods are normally used (Kong et al., 2015b).

Region growing: Region growing techniques are relied on the sub region features, connectivity, and structural characteristics of object (Luo et al., 1998). Region growing methods divided into two steps, first: the selection of sub regions (or seed) that have uniform or near uniform properties and second: categorization of neighboring pixels to specify if pixels should be merged into the larger sub region or not by a cost function minimizing (Irshad et al., 2014). Splitting and merging strategies can improve the region growing techniques, which both region-splitting and also region-growing can occur. A region can divide into or combine

with others depending on the number of predefined criteria (Luo et al., 1998).

Active Contour Models (ACM): Active Contour Models (or Deformable models) are elastic curves that can be applied to capture object edges evolve on the images utilizing gradient information by an energy optimization method. Generally in image culture, the energy is the function of the image's features. The energy function is the sum of its internal energy and external energy of the active contour which internal energy controls the deformations made to the active contour and external energy controls the fitting of the contour onto the image. The optimization is a technique to minimize the energy function from an internal and external energy.

Therefore, in object segmentation, the points which assign minimum energy levels, make the object boundaries. The energy level is defined as the intensity of the image that can be used to detect corners and terminations in a slightly smoothed image. The energy functions are frequently clarifying curve shape discontinuity in the gray levels along the contours (Irshad et al., 2014). Active contour models are able to be formulated either obviously, as like a parametric contour (2D) or surface (3D), as the zero-level of a function with dimensionality (n-D) one higher than the image to be segmented (Meijering, 2012).

The main advantage of active contour models is the capability of giving a linear determination of the object shape and the additional processing is not essential at the convergence time. One of the limitations of this model is required to find strong image gradients to discover the contour that can be the reason of over and under segmentation which accountable for decreasing the correctness of the analysis, and also in due to frames with low contrast. These models broadly applied to tracking cells, immune cells, cancer cells, amoeba, and fibroblasts has been successfully feasible. (Svensson et al., 2018; Huth et al., 2010). This algorithm fails when objects undergo enough large displacements (Huth et al., 2010) or its overlap is low between consecutive frames. Moreover, in detecting extant cells in the field of view. In Li et al. (2008b), Li and co-workers, by applying local association scheme, treated new cells. Although, recent external-force-effected procedures are incapable in dealing with low-contrast edges and suffering from the effect of initialization sensitivity. Therefore, in Huang and Liu (2015) Yali Huang et al. proposed a functioning contour model for segmenting and tracking the exact membrane of each lymphocyte in phase-contrast microscopic images.

There are various other methods that can be utilized for segmentation. For example, once the positions of cell have been detected, they must be connected during the time in order to form cell trajectories. The simplest solution to execute this function is to connect each segmented cell in a frame to the nearest one in the next side frame. In this case, 'nearest' may mention the spatial location and cell centroids possibly used as points of reference. This strategy will respond on condition that cells move slowly according to the chosen frame-rate and are not compact. Feature matching is another method with better results. This procedure locates homogeneous cells utilizing an expanded list of features like volume, morphology, surface and curvature which expand the conception of distance further than the spatial location. Basically these techniques need user to indicate the maximal distance in which cells can transfer through each couple sequential timeframes. These methods are based on a correct segmentation toward keeping the accurate matches as possible. Furthermore, each morphology change, through sequential timeframes effect on single cell causes one cell becoming two different cells, producing broken trajectories (Masuzzo et al., 2016). Additionally, there is another method termed gradient flow that is a new idea for 3D cell segmentation. This method relies on the supposition that the gradient vectors generally point to the center of the cell. However, this supposition only happens in a way that required parameters for elastic deformation are exactly selected. The last results are subject to distribute for bimodal voxel intensity within gradient-driven volume growing. In practice, one of the serious challenges in this method is choosing the regulation methods of gradient field and identify cell component in every sub-volume (Kong et al., 2015b). Therefore, to address these problems, Kong et al. (2015a) presented a cell segmentation method which can recognize each cell of its 3D fluorescence microscopic image. Gilad et al. (2018) introduced a method for both mitosis event detection and mother-daughter connection with data of fluorescence microscopy. However, their method was not satisfied in the condition that cells' boundaries cannot be easily distinguished. In example of deep learning assist in tracking, Falk et al. (2019) provided (U-Net) a package based on deep learning approaches in order of detecting and segmentation. It is an image plugin used to help non-machine-learning specialists to analyze their information by means of U-Net on local computers or remote server/cloud service. In another work, Boukari and Makrogiannis (2018a) introduced a method for cell tracking, lineage construction, and quantification.

Yang et al. (2016) developed an automatic algorithm named "MCDT" (Migrating Cell Detector and Tracker) for the merged function of migrating cell detection, segmentation and tracking of vivo fluorescence time-lapse microscopy imaging information. Coudray et al. (2018) instructed a deep convolutional neural network (origination v3) on whole-slide images to automatically category them into LUAD, LUSC or lung tissue that can be used as an aid for diagnosing lung cancer by means of histopathology slides. While great improvement has been occurred in detecting cells with high contrast and predictable shape, but in different conditions all properties do not exist all together. Therefore, most of the interesting potential researches-for example transferal patterns of astrocytes to scratch wounds-have been reduced of qualitative aspect in nature. Liu et al. (2017) built a model for plant cell matching using the cells' 3D spatiotemporal context. They combined the 3D local graph matching model with the Interacting Multi-Model filter (IMM) to track the population of plant cells. In another study to find plant cell matching, Liu et al. (2019a) proposed a model by combining convolusional neural network and local graph matching. Their model applies cell pairs similarity information and cells spatial–temporal contextual information.

Nejati Javaremi et al. (2013) introduced an algorithm in which cell movements dynamics for formulation was considered and parametric active contours was used for automating segmentation and tracking the external shape of the cells in microscopic images.

2.2. Tracking by model evaluation

In these approaches the representations of object appearances or shapes are developed from analyzing of each frames or in spatiotemporal volumes in order to keep track of moving objects over time. Briefly, it searches the relationship of objects in each frame to find similarities (Nketia et al., 2017). It is used for situations where the object is not static and its feature changes at any moment. Model evolution approach obtains the edge of an object in the frame by developing the contour of the former frame. In this examination, it contains initialized the object segmentation in the first frame and refreshing at next frames. The outcome of any frame is applied as a beginning requirement for examining the next frame (Nketia et al., 2017; Masuzzo et al., 2016). These models are broadly applied in applications like object tracking, edge detection, shape modeling, segmentation, finding the mobility of the active objects, and handle the

cell division by providing a primary segmentation for the first frame (Sacan et al., 2008b). Tracking by model evaluation faces two challenges. First, incorrect segmentation in due to the low-contrast edges or noise pollution. Second, sensitivity of models to the beginning position (Huang and Liu, 2015). Generally, tracking by model evaluation divided into two types by the representation and implementation: the parametric and the geometric active contours.

Parametric models establish energy function contained internal and external energy condition. These conditions are constraint forces introduced by the users. They are determined clearly as parametric contours and also are often used to 2D applications. Therefore, parametric models provide a speedy and real-time execution except topological events like track dividing or combining. Parametric models are capable of producing better estimates of cell morphologies, and are used to track single object and partial occlusion of objects (Mean SLDUA, 0000). However, they must be able to control cell–cell contacts and mitosis (Nketia et al., 2017; Masuzzo et al., 2016). These methods are nearly simple and optimal in terms of computational cost (Meunier et al., 2010).

On the other hand, **non-parametric** deformable models display, implicitly as a level set of a scalar function with higher-dimensional that is able to tackle the changes of topologies at a higher cost of computational complicatedly (Kalaidzidis, 2007a). However, in these methods re-initialization is necessary in the conditions of cell appearance or disappearance from the field of view or in the same case of quick movement depending on the chosen frame-rate (Nketia et al., 2017; Masuzzo et al., 2016; Huang and Liu, 2015).

Although, there are several differences between parametric deformable models and non-parametric deformable models, but they share an implicit principle: First starts with a difficult segmentation, the models repetitious developed to minimize a particular energy function (Nketia et al., 2017; Masuzzo et al., 2016). It must be noted that, if alters in collecting information of the object; consequently, in these terms the tracking is not reliable (Meunier et al., 2010). These models are appropriate in tracking of single objects, physical initialization, multiple images, fully and partial occlusion of objects, object motion by translation, scaling and rotation (Mean SLDUA, 0000; Huang and Liu, 2015).

2.3. Tracking by filtering

Tracking based on filtering approaches (also known as sequential Monte Carlo techniques (Liu, 2008)) often employs stochastic filtering techniques based on the implicit movement or appearance model of tracked object. In this case, stochastic filtering methods are based on the basic movement or appearance pattern of the tracked object. Therefore, there is expectation in this model in which follows some assumptive motion pattern, but does not need an exact segmentation of the tracked objects. As a result, these methods are suitable for large displacements (Nketia et al., 2017) and broadly used in multiple-object tracking systems. Filtering methods are based on a principal idea that estimate the density function of the state posterior of the object through a collection of random particles with connecting weights. Filtering methods divided into three stages: selection, prediction, and measurement, which are performed iteratively. The selection step includes generating a new set of particle by choosing them with the highest probability of posterior among the prior particle set. The prognostication step, modifying each particle considered state model of the surrounding region-of-interest, while in following measurement step the weight of every particle is reappointed relied on the new information. It should be noted that a resampling step is often necessary, to prevent the agglomeration of weight by good particles and raised punishment of bad particles. Eventually, summing the weight of particles that give an estimation of target distribution and applying the last state estimating in order of initialize the next frame tracking. Although, a large number of proposed particle filters in a series of video sequences, tracking is based on a particular feature, but multiple-feature tracking supplies a better explanation of object, eventually enhances the robustness of this method. In most recent particle filtering methods, human communication or other sort of initialization is essential to explain the prior explanation of object states in the initial frame. Moreover, it is necessary that the deformation of the tracked object remains small, because large deformations preclude their application to individual cell tracking (Nketia et al., 2017; Masuzzo et al., 2016; Smal et al., 2008b). Filtering tracking approaches can be categorized into deterministic and statistical methods (Meunier et al., 2010):

Deterministic Method: The deterministic method can be applied to associate the new situation and the previous situation when limitation of regarding the movement of the object can be certain (Meunier et al., 2010). This method of creation a communication between each object in the prior frame with the single object in the present frame, works on qualitative movement heuristics, which is accomplished with the aid of a collection of motion limitation. (Samdurkar et al., 2017b)

Statistical methods: Also called probabilistic methods. These methods are applied when the state space approach is more efficient for determining attributes such as location, speed and acceleration (Meunier et al., 2010). Probabilistic methods consider the object's measurement and indefiniteness to assign the communication. They solve the tracking problems same as the noise exists in the measurements. (Samdurkar et al., 2017b)

In Godinez and Rohr (2015), Godinez WJ et al. presented an approach in order to track multiple particles relied on probabilistic data connection. In another work (Godinez et al., 2007), to explore the spatiotemporal relationships of viruses over time, W. J. Godinez et al. originated deterministic and probabilistic methods in order of multiple virus tracking. Moreover, Hodneland et al. (2016) introduced a consecutive approach that develops a partial differential equation, which is relied on an immersion front geometric description.

3. Tracking tools

Intuitive and supportive computational tools are an important factor of providing end users access to some tracking technology (Nketia et al., 2017). There are two categories of platform for analyzing biological images: **commercial platforms** (e.g. ImarisTrack, ImagePro Plus, Velocity, Amira, MetaMorph,) often concentrate on ease-of-use. They broadly cover tasks of the image processing, and are suitable for users with low level of expertise. Basically, the major facts of image processing algorithms are unapproachable, which is unsuitable for solving frontier research questions. While, in **open source platforms** (e.g. CellProfiler Xu et al., 2019, Vaa3D, BioImageXD Carpenter et al., 2006, Vaa3D, BioImageXD Kankaanpaa et al., 2012, Icy de Chaumont et al., 2012, KNIME, FARSIGHT Bjornsson et al., 2008, and ImageJ Abràmoff et al., 2004, Fiji Schindelin et al., 2012 and etc.) algorithm details are completely transparent, and often target the expert user. Open source tools give users the ability to make new inventions without restriction. These tools are better chosen in some situation where some factors such as security and support are less important than low-costs. Here, in Table 2 we mentioned some available computational tools in the terms of cell tracking approaches.

ImageJ is one of the oldest and the most powerful tools for scientific image analyzing. ImageJ was created by Abràmoff,

Table 2
An overview of a selected set of well-known computational cell tracking tools.

Tool	Task	Platform	Automation	Availability	URL
CellTrack (Sacan et al., 2008b)	Number of cells, boundary positions, cell speeds, path length, cell area, cell deformation.	Win	Auto	Free	http://db.cse.ohio-state.edu/CellTrack/
CellTracker (Shen et al., 2006)	Number of cells, cell centroid positions, cell surface areas (cytoplasm and nucleus individually), cell displacements, cell velocities.	Win	Semi	Free	http://go.warwick.ac.uk/bretschneider/
MetaMorph [Molecular Devices, USA]	Cell displacements, cell angles, cell velocities	Win	Auto	Commercial	http://www.moleculardevices.com/Products/Software.html
Diatrack [Semassopht, Switzerland]	Particle detection, particle selection and particle tracking, analyze moving particles, extract quantitative information	Win	Auto	Commercial	http://diatrack.org/
ImarisTrack [Bitplane, Switzerland]	Number of cells, cell centroid positions, cell surface areas (2D), cell volumes (3D), cell displacements, cell velocities, path curvature, number of fusions of tracks, number of divisions of cell track	Win/Mac	Auto	Commercial	https://omictools.com/imaristrack-tool
Particle Tracker (Sbalzarini and Koumoutsakos, 2005)	Particle trajectories, image segmentation, interaction analysis	ImageJ	Auto	Free	http://weeman.inf.ethz.ch/ParticleTracker/
ManualTracking [Fabrice Cordelie`res]	Cell centroid positions	ICY; cross platform	Manual	Free	http://rsb.info.nih.gov/ij/plugins/track/track.thtml
iTrack4U (Cordelières et al., 2013)	Single cell tracking, speed, acceleration, angle, persistence, statistics on computed features	Cross-platform	Auto	Free	https://omictools.com/itrack4u-tool
LineageTracker (Downey et al., 2011)	Nucleus areas, cell shapes, cell sizes, cell lineages	Cross-platform	Semi	Free	http://go.warwick.ac.uk/bretschneider/
u-track (Jaqaman et al., 2008)	Cell tracks merging, cell tracks splitting, cell displacements	MATLAB; Win	Auto	Free	http://lccb.hms.harvard.edu/software.html
MtrackJ (Meijering et al., 2012a)	Cell centroid positions, cell displacements, cell velocities, cell angular changes	ImageJ/Fij; cross platform	Manual	Free	http://rsb.info.nih.gov/ij/plugins/track/track.thtml
TrackMate (Jaqaman et al., 2008)	Cell displacements, cell velocities, cell orientations, trajectory length, trajectory displacement (enables manual editing of tracks)	ImageJ/Fij: cross platform	Auto	Free	http://fiji.sc/TrackMate
Volicity [Perkin Elmer, USA]	Number of cells, cell centroid positions, cell surface areas (2D), cell volumes (3D), cell displacements, cell velocities	Win/Mac	Auto	Commercial	http://cellularimaging.perkinelmer.com/products/volocity/demo/
SpotTracking (Chenouard et al., 2013)	Cell displacements, cell velocities, trajectories editing	ICY; cross platform	Auto	Free	http://www.bigwww.epfl.ch/sage/soft/spottracker
ClusterTrack (Matov et al., 2010)	Intracellular microtubule dynamics, spatial patterns	Matlab	Auto	Free	http://lccb.hms.harvard.edu/software.html
DcellIQ (Li et al., 2010)	Quantitative, reproducible and accurate interpretation of cell dynamic behaviors	Matlab	Auto	Free	http://www.cbi-tmhs.org/Dcelliq/
CellProfiler (Carpenter et al., 2006)	Cell displacements, cell velocities, cell morphology, area, perimeter, solidity, form factor, eccentricity	Cross-platform	Auto	Free	http://www.cellprofiler.org/
FARSIGHT (Bjornsson et al., 2008)	Automated 3D segmentation, analyzing three-dimensional, delineate and classify cell nuclei, compute quantitative associations, and cell processes	Cross-platform	Auto	Free	http://www.farsighttoolkit.og/
plusTipTracker [Danuser lab]	Microtubules dynamics tracking, data visualization, statistical processing of parameters of microtubule dynamics	Matlab	Auto	Free	http://lccb.hms.harvard.edu/software.html
DeepSeed					

M.D. and co-workers that supporting image processing functions (e.g. mathematical and legitimate performance in images, contrast control and managing, smoothing, sharpening, Fourier analysis, median filtering, and edge detection). Vast acceptance and expandable plugin architecture has made it a strong implement of choice from a broad fields of study for scientists. Next, Schindelin et al. (2012) proposed Fiji, an open source software tool, which brings up to date the underlying architecture of imageJ. This tool enables researchers to concentrate on the process of making inventive answers for biological image analysis (Masuzzo et al., 2016; Schindelin et al., 2012). There are some generic packages which although, are not adjusted to support biological simulation per se, but can be used in biological modeling such as Mathematica or MATLAB (Masuzzo et al., 2016).

More specialized solutions are also available, for example AceTree (Boyle et al., 2006; Murray et al., 2006), and ALES (Braun et al., 2003) are designed for the visualization, creation, and analysis of the cell lineages. StarryNite (Murray et al., 2006) has been created specifically for the study of embryogenesis that in addition to segmentation and individual cell tracking, controlling all of the cell divisions and rebuild of the thorough cell lineage tree are needed. Meijering et al. (2012b) LEVER 3-D combines quantitative and visualization analysis of 5D and big composition of confocal fluorescence microscopic images. It is automatically segmented, tracks, and lineages of sequential data image and next lets the user observe and edit the outcomes of algorithms in a stereoscopic 3D screen concurrently observing the stem cell linage tree in a 2D screen (Wait et al., 2014). Ilastic platform is a user-friendly and easy implement in use of connecting the image assortment, segmentation and analysis. In particular, it prepares accessibility to some kind of professional machine learning on the basis of algorithms for segmentation and tracking (Nketia et al., 2017). Diatrackit offers a strong new approach to analyze moving particles movies and based on an innovative graphical user interface accessible particle tracking to microscopists and biologists. From image pre-processing of particle detection, particle selection and particle tracking, tracking functions are arranged from top to bottom (Vallotton et al., 2017). Tri-track has good performance in complex dynamics such like dense antiparallel flows, or in situations where object dislocations are notable. It is effective in the presence of deterministic dynamics (Vallotton and Olivier, 2013). SpatTrack combines several algorithms for particle tracking and analysis of 2D cell images. Graphical user interfaces (GUIs) are executed for every subroutines and joint to the original program of tracking. It is carried out dependable vesicle tracking and fitting the mean square dislocation of particles to different diffusion/transport models. Moreover, without any programming knowledge prepares the feasibility to analyze tracked intracellular particles in a several ways. Lund et al. (2014). It is free for download.

3.1. Intracellular organelles

The regular functions of the cells for their elemental effect in living organisms are the result of a composite intracellular and intra nuclear processes, containing thousands of proteins and other cell structures. For instance, live cell imaging populations over some days have proved that signaling dynamics can manage cell reactions. Therefore, having computational tools can increase the efficiency of cellular analysis with exact results in order to ease a deeper understanding the connection of the changes of molecular and cellular with neuronal maturity (Shahen et al., 2018). Observations of over several days live cell imaging showed signaling dynamics has an important role on controlling of cell fate decisions (Cooper et al., 2017).

Embryonic morphogenesis includes tissue deformation and development, which frequently happens quickly over brief timescales. Tissue deformation appears by means of local cellular activities, contains cell manipulation, morphology and/or size changes, and cell reorganization. Somites are paired embryonic segments forming through vertebrate development from unsegment mesoderm in a regular order. Their transient structures are so important because they produce musculoskeletal system cell lineages in the trunk. Like skeletal muscle, endothelial cells, bone, tendon, and cartilage. However, the literatures about cellular dynamics based on the morphological change through somite differentiation is very little. In this way, McColl et al. (2018) inspected cellular reorganization and morphogenesis in separating somite utilizing alive multi-photon image of transgenic chick embryos, express a membrane-bound GEP. They tried to describe and quantitate the behavior of single cell in real time, alive, growing somite. They explicitly focused on the dynamic changes of cellular in two principal parts in somite, suited to the middle or suited to the sides, to research broad morphological changes.

Mitochondria is providing the neurons energy requirements. The correct localization is important and guaranteed by means of the system of transportation. Generally, in neurons mitochondrial trafficking is especially important, on the grounds in order to facilitate neuronal function, mitochondria should leave the soma and move through the axon and network of dendritic. Abnormal mitochondrial trafficking has been expressed in various neurological disturbances. Consequently, deep understanding of fundamental mechanisms of mitochondrial trafficking through neuronal maturity is essential to our knowledge about pathogenic changes based on neuron-degeneration and other neurological disturbances.

There are some available macros used as an aid for performing different analyses of mitochondria, for example MiNA (Valente et al., 2017) and KymoAnalyzer (Neumann et al., 2017). However, there was a necessity for each macro to execute both suitable mitochondrial trafficking analyses simultaneously by counting mitochondria in a semi-automated manner. Thus, MiTrakD (Shahen et al., 2018) utilized existing imageJ implements for fast semi-automating, and mitochondrial trafficking analysis and distribution, containing abundance, velocity, traveling distances and localization in neurons.

There are some situations that can be included as segmentation mistakes, for example, combining border cells incorrectly or preventing valid cell insurances because of the poor estimation with bounding boxes. To address these, Schmidt et al. (2018) proposed star-convex polygons to localize cell nuclei. As another example, most of the cell tracking software packages are based on ‘frame to-frame linking’ nuclei or cells segmentation (Meijering et al., 2012b). In these techniques, errors in segmentation can seriously compromise tracking outcomes. Hence, Cooper et al. (2017) presented NucliTrack a tool for automatic segmentation, feature extraction and tracking from nucleus labeling with fluorescent. The detection and segmentation of fluorescently labeled objects are based on marker controlled watershed. Also, the interface contains implements for inspecting, correcting and exporting data on track nuclei.

Moreover, the discovery of nuclei of cell is a challenging subject in due to limitations of quality of cellular image and nuclear morphology diversity, such as various nuclei shapes, sizes, and coverage multiple cell nuclei. Hence, to cell nucleus detection, Tofighi et al. (2019) provided a network structure (SP-CNN) in which can include ‘expected behavior’ of nucleus forms through two parts: learnable layers execute the nucleus detection and a firm processing part in which leads the learning with former information.

3.2. Cellular process and events

Functional cell movement is essential for cell living. Cell movement is required for a number of activities performed within the body, such as the formation of the body of a living organism, tissues, organs, and cell shape. If the cells have not the ability of movement, they cannot grow, to be divided or migrate to the areas that need. Cell migration has an essential role in physiological phenomena such as neural growth, cell cycle, wound healing, mental retardation, and immune function, also in some disorders like as neurological diseases, fibrosis and metastasis of cancer (Masuzzo et al., 2016). For more information about cell migration see Majumdar et al. (2019), Vicente-Manzanares and Horwitz (2011), Aman and Piotrowski (2010), Reig et al. (2014), Li et al. (2013) and Horigane et al. (2019).

Long-term observation can alter normal biology. There is a limitation of tools that track movements in vivo without perturbation. As a biological example, we can point to the Drosophila follicle (egg chamber). Epithelium-encased organ grows in adult female venter, and cell migration of whole-tissue collective extends several consecutive days. This migration leading to gyrations of the whole organ through several cycles, and has been recommended to drive its length from spherical to ellipsoid. Follicle rotation and its dynamics (as due to, its long-term nature and internal site) have been imaged and determine quantity through exploits in ex vivo culture. It has stayed unknown, if the dynamics of cell migration changed in vivo. In this way, Chen et al. (2017), developed an implement called M-TRAIL (matrix-labeling technique for real-time and inferred location) which reveals histories of migration in fixed tissues.

Neural stem cell (NSC) is another notable challenge for time-lapse tracking analysis because of the existence of high motility, photosensitivity, and growth at confluent densities. There are several studies that have been extensively on NSC migration using time-lapse imaging (Noctor et al., 2004; Yokose et al., 2011; Namba et al., 2011; Sakaue-Sawano et al., 2008). In Roccio et al. (2013) and Daynac et al. (2015) researched cell cycle by means of single-cell tracking of Fucci-cell cycle reporter showing mouse NSC. Other works (Blong et al., 2010; Keenan et al., 2010), have searched each cell tracking in time-lapse imaging of human NSC. These investigations were focused on short term experiments (about 24–72 h). However, Piltti et al. (2018) presented a way to analyze the dynamics of cell cycle, migration, and lineage choice in cultured multi potent mouse or human NSCs utilizing single-cell for long term (about 7–14 days) analysis of live cell.

Generally, in microscopic live cell images, the cells are thickly spread, move quickly and are often the same in appearance. In practice, it is unable to track manually hundreds of similar cells by means of their movement of captured movement in a series of image sequences. Therefore, computational tracking methods can automatically detect and track multiple moving cells in the shortest possible time. There are several studies have been investigated on motility tracking (Smal et al., 2008b; Thomann et al., 2002; Altinok et al., 2007; Gerlich et al., 2003; Ong and Verma, 2019; Xu et al., 2019; Brasch et al., 2019).

Sometimes targets are affected by the proximity and/or effect of the other targets. In subcellular dynamic tracking, the identification of objects throughout a communication, decreasing intensely tracker failures. For this purpose Smal et al. (2008a) and Khan et al. (2004) objected to track the pattern of moving of various tens of targets communicating with each other recommended similar tracking method rely on the particle filtering and MRF priors. Tokunaga et al. (2014) provided a method that was an extension of Smal et al. (2008a) and Khan et al. (2004) to develop the tracking deed by using the spatiotemporal variation of cells with motion. Their recommended methods were relying on the kernel density estimation strategy (KDE) (Silverman, 2018; Wand and Jones, 1994) and several optimization modules. In Hirose et al. (2018), Hirose and co-workers presented a multi-cell tracking technique that was rely on a MRF model and effective technique to address the task of tracking hundreds of cells.

4. Cell tracking data sets

Depending on the types of cells, to perform computer analysis on them, the main and most important step is to find and collect images data from the object of interest. To the best of our knowledge, there are two main repositories for images datasets: Cell Tracking challenge (Ulman et al., 2017) and Cell Tracking Library. The Cell Tracking Challenge consists of 2D and 3D time lapse video sequences of fluorescent counterstained cells or nuclei moving on a flat substrate, along with 2D phase contrast, bright field, and differential interference contrast microscopy videos. The videos cover a broad range of quality and cell types (e.g. noise levels, spatial and temporal resolution and so on). Moreover, Cell Tracking Challenge provides videos (2D and 3D) of synthetic fluorescently stained nuclei and cells of different forms and motion patterns. All 2D and 3D datasets are freely accessible for download in http://celltrackingchallenge.net/. The Cell Image Library is a repository for cells' images and movies of a variety of organisms, cell types, and cellular processes which are acquired from literature publications, historical and modern collections, and recruitment. This database represents cellular functions and architecture with high quality images, animations, and videos. This Library is a public resource for research on cellular activity and a tool for education. The images and datasets of this repository are freely available in http://www.cellimagelibrary.org/pages/datasets.

5. Discussion, challenges, and future directions

Today, computational techniques are widely used in various fields of biology and medicine (Emami et al., 2020; Ferdousi et al., 2020; Raef and Ferdousi, 2019; Raef et al., 2020). Computational approaches and Computational object tracking techniques offer the possibility of identifying of cellular processes, cellular organs (e.g. plasma membrane, cytoplasm, and nucleus), and monitoring the motility in the consecutive frames. Finding barriers in the path to avoid overlapping, monitor the abnormality and any defect in cell maturity cycle are some applications of cell tracking. Tracking methods have acceptable results in some subcellular problems that high level of spatial resolution is important (e.g. the molecular dynamics of dendritic spines). Sub-cellular tracking techniques have a significant performance in the problems containing messy nonstoichiometric processes, like protein oligomerization and clustering.

On the other hand, there are many challenges in computational object tracking that addressing them is required important developments. One of the main challenges that have to be considered is errors in tracking results. According to the outcomes of Yan et al. (2019), Gupta et al. (2019), Buonviri et al. (2019), Liu et al. (2019b), Yoo and Lee (2019), Nikouei et al. (2019), Meirovitch et al. (2019), Sun et al. (2019), Fiaz et al. (2019), Gupta and Kulkarni (2008), Li et al. (2017) and Sigal et al. (2004) studies, we have classified the key factors that affect the tracking results and causing errors. (See Table 3)

Noises: It is difficult to detect objects when images have a high noise ratio and in some cases, the noise is followed instead of the objects. This occurs when the noise is almost as high as the intensity of the objects. Although, several strategies have been proposed for noise removal, but, in the most noise removal methods, to remove more noises, they sacrifice a lot of real details that sometimes are useful. Therefore, there is a need for more efficient methods that sacrifice less information. Moreover, still need for more researches in this scope to study why noise is generated and how noise affects the tracking by the results of the experiments that it is an effect. **Background subtraction problems:** In the most of tracking algorithms, very dark pixels of objects are classified as either background or shadows. Additionally, when multiple shadows added together or are on a very dark background will not be detected efficiently. Therefore, as a next step, the need for new studies about developing a new system that can handle these problems is felt.

Difficulty in reference input information: Generally, the most tracking methods are using two sources for detection and tracking of objects: visual features and motility information. According

Table 3
This table shows the number of the challenges which object tracking in image processing met to them. On the above described their types and methods that used as solutions in video object tracking.

Name	Definition	Types	Solutions
Noise	The quality of the images decreases by noises. It may happen during capturing, transmission, etc. Noise removal is a serious and main task in image processing.	• Film grain • Uniform noise • Shot noise • Salt pepper noise • Gaussian noise • Anisotropic noise • Periodic noise, and etc.	✓ Filtering ✓ Block matching algorithm ✓ Random field ✓ Wavelet transform ✓ Statistical method ✓ Non-local means ✓ Anisotropic diffusion ✓ Deep learning
Occlusions	Occlusion occur when an object of interested tracking is hidden (occluded) by another object while the object is present at the scene yet. Generally, it is happened in three conditions: self-occlusion, inter-object occlusion and occlusion by the background scene structure.	• Non-occlusion • Partial occlusion • Full occlusion • Long term full occlusion • Linear dynamic models • Nonlinear dynamics	✓ Depth analysis ✓ Fusion methods ✓ Optimal camera placement method ✓ In the surveillance scene, installing more cameras ✓ Dynamic graph
Difficult object motion	It is such as an irregular dynamic (such so fast-moving or so slow-moving), changes in the scene in the moving regions. Generally it is hard to recognize the related pixels of some types of moving objects.	• Prismatic links • Rotational links • Helical links • Very (fast or slow) movement	✓ Temporal differencing ✓ Moving camera by temporal differencing method ✓ Statistical Approaches ✓ Optical Flow ✓ Background Subtraction in static scene
Complex objects structures	The problem of explaining and detection categories of objects (e.g. cells, nucleus, etc.) is central to computer vision. It means not knowing how many features one must use and how these features detected and represented.	• Spatial relationship between parts of an object • Visual features • Specific appearance and shape • Spatial limit and extent • Position (.etc.)	✓ Graphical Object Models ✓ Unified framework with combining object tracking and object detection ✓ Dynamic graph
From 3D to 2D world	When the 3-D image is converted into 2D, a lot information is lost.	–	✓ Automated single camera object ✓ Position of the camera
Background subtraction problems	Sometimes in an image's regions of interest, objects (cell, etc.) are in its background. If detection based on this scope it may has some challenges.	• Illumination changes • Dynamic background • Clutter • Camouflage • Presence of shadows • Motion of the camera • Bootstrapping • Challenging weather Speed of the moving objects and intermittent object motion	✓ Filtering ✓ Robust cameras ✓ Background Subtraction in static scene

to recent advances in deep learning techniques, it seems that by using these methods, which system learns that how to deal in the face of any of these movements, the accuracy of tracking systems has significantly increased. It should be noted that, this goal is not possible except by creating a large dataset including various types of complex movement (e.g. prismatic, rotational, helical, very fast /slow movements) that are constantly updated. Other problems in temporal object tracking are the detection of objects which their shape, size, and appearance change considerably in different times. Some works still need to be accomplished in how many features must use in the first and how these features are found and exposed.

Imperfect and entire object occlusions: Occlusion handling is an unavoidable difficulty in object tracking. Therefore, further works need about the tracker function during the occlusion occurring and after the occlusion.

Information loss: converting 3D to 2D images leads to information loss. Performing further studies to reduce the effects of this conversion based on new methods by using automated cameras for target object or calculation of the correct position of the camera can be helpful.

Finally, the run-time and speed optimization of the algorithms are a classical issue. In the most tracking algorithms, with increasing calculation their speeds are reduced and, as a result, run-time is increasing. Additionally, most of these methods do not depict the cellular environment. Therefore, designing multi-tasking systems that can track multiple objects, and simultaneously visualize the cell's environment is urgently needed in many biological problems, which can be helpful for analyzing complex cellular events. (E.g. endocytosis and exocytosis).

Moreover, due to the recent advances and remarkable performances of machine learning algorithms and deep learning approaches in bio image analyzing, research on cell tracking is likely to continue with the help of these methods (individually or in combination) which can be applied at all stages of cell tracking. Consequently, it leads to new opportunities and challenges in this scope.

Although, in the last two decades, a variety of object tracking algorithms have been implemented, however, in biological studies only a few of them have been used and most of them are not yet fully exploited. It is expected that further researches on the combination of in vitro experiments with in silico methods create new insights in this field.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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